

Seonghwan Lim

Computer Vision Engineer

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SUMMARY

Computer vision engineer with an M.S. in Computer Science. I build detection, tracking, pose estimation, and image-measurement systems end to end, from dataset design and labeling through training, benchmarking against independent references, and real-time inference.

EDUCATION

Illinois Institute of Technology — M.S. Computer Science, GPA 3.9/4.0 May 2026

Seoul Nat'l University of Science and Technology — B.S. Electrical and Information Eng., GPA 3.6/4.0 Aug 2023

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, Bash

Computer Vision: Object detection, classification, multi-object tracking, 2D/3D pose estimation, image processing, OpenCV, YOLO

Machine Learning: PyTorch, CNNs, Vision Transformers, multi-task learning, data augmentation, scikit-learn, XGBoost

Data and Tools: Dataset design, labeling QC, NumPy, Pandas, Git, Linux, Docker, CUDA real-time inference, embedded systems

EXPERIENCE

Camera-Only Worker Safety Monitoring Jan 2025 – Present

Research Assistant, iConSenSe Lab, Illinois Institute of Technology

- Built a camera-only worker monitoring system on ordinary site video, no wearables. A YOLO pose model extracts keypoints, ByteTrack holds worker identity across frames, and downstream heads roll activity, work state, and posture into a 10-second exposure score at a median 23 fps for 3 to 9 workers on an RTX 2060.
- Checked every module against an independent reference across 42 subjects, reaching 0.861 activity accuracy, 0.775 work-state F1, and 0.793 correlation with a 3D-pose ergonomic reference. Scoped workload as an exposure indicator rather than a fatigue measurement after it reached only 0.526 against indirect calorimetry.
- Built the CO₂ sensing rig on Raspberry Pi for the field cross-check, and separately lifted 2D poses to 3D with MotionBERT across 61 HoloLens 2 sessions. Presented at the BuiltWorlds Global Summit and received a \$12,500 research grant.

Fine-Grained Road Surface Classifier Jan 2025 – Jul 2026

Research Assistant, MMCom Lab, Illinois Institute of Technology

- Trained a classifier over 27 road-surface types on RSCD, roughly 960k training images against a 50k test split. Traced most errors to a long-tailed class distribution and to surface types that look nearly identical in a single frame.
- Fixed the data before the model, filling out rare classes with diffusion-generated images, then tried multi-task and mixture-of-experts heads to separate the look-alike classes.
- Pushed EfficientNet-B0 from 91.70% to 93.33% over a standard-augmentation baseline.

Image-Based Process Control for Microfluidic Mixing Jan 2025 – May 2026

Research Assistant, Bhushan Research Group, Illinois Institute of Technology

- Wrote the measurement code that registers and normalizes CFD contour and experimental fluorescence images, then computes a 0 to 1 mixing index from outlet pixel intensity.
- Benchmarked SVR, MLP, XGBoost, Random Forest, and linear regression as CFD surrogates. SVR won on held-out data with MAE 0.0014 and R² 0.9998, close enough to stand in for a full CFD solve during inverse screening for inlet conditions that hit a target mixing value, which I then confirmed with fresh CFD runs and six fluorescence experiments.

Two-Stage Parcel Inventory Recognition Jan 2023 – Dec 2023

Undergraduate Researcher, VCL Lab, Seoul National University of Science and Technology

- Built two-stage recognition for photos of stacked parcels. DINO and YOLO detect product regions and Weighted Boxes Fusion merges them into cleaner boxes than either alone, then EfficientNet-B7 with test-time augmentation classifies each crop.
- Placed 1st of 19 teams, identifying 56.6% of the 53 evaluation products against 49.1% for the runner-up, at 0.80 mean IoU, the only team to pair the top recognition rate with localization above 0.75 IoU.

SELECTED PROJECTS

Road Damage Analysis (github) — Pipeline that pulls road geometry from OpenStreetMap, downloads the matching Street View frames, detects pavement damage with YOLO, and renders an interactive map with clustering and a heatmap for inspection triage.

Marionette Motion Control (writeup) — Kinect RGB-D 3D joint capture driving a C++ servo control layer on an embedded controller, each servo clamped to the puppet's reachable range so a bad pose estimate cannot damage the rig.

AWARDS AND PRESENTATIONS

Research Grant, BuiltWorlds Global Summit (\$12,500) 2026

1st Prize, CJ Logistics Future Technology Challenge (\$15,000) 2023

Sensor-Free Wearable System for Activity and Fatigue Detection in Worker Safety Monitoring, BuiltWorlds Global Summit 2026

Real-Time Control of Microfluidic Mixing via Machine Learning-Driven Prediction, BMES Annual Meeting 2025

Development of Marionette Motion Control System, ICROS-KROS Conference 2022